Review of Carbon Dioxide Removal Technologies

0. Abstract

Climate change poses a greater threat to the entire globe than any other humans have created. Left unchecked, its devastating effects will extend to threaten not just humans, but all life. Mitigation plans center around decreasing greenhouse gas emissions but also require another strategy. Enter carbon dioxide removal (CDR) technologies. While still largely untested, they remove existing and offset future emissions, and in some cases have just begun to reach commercial deployment. This review aims to shed light on some of the more developed CDR technologies and the facets of that make each more or less promising. Focusing on point source capture, direct air capture (DAC), and direct ocean capture (DOC), this paper provides an overview of the state of the young technology today.

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Pretending this is a literature review in a real journal

1. Introduction

For millennia, humanity has been making relatively small, permanent changes to the environment through things like extinction hunting and dam-building. However, it was the arrival of the industrial revolution that jumpstarted a whole new scale of irreparable and unmistakable damage to the entire planet. Greenhouse gases, particularly carbon dioxide (CO_2), have been building up in the atmosphere and oceans causing increases in things like storms, fires, biodiversity loss, extreme heat, and respiratory illnesses [1]. Measurements and extrapolation like those in Figure 1 show a clear warming trend since 1880 through the greenhouse gas effect totalling 1°C.

At the 2013 conference of parties of the United Nations Framework Convention, also known as the Paris Climate Accords, researchers presented the Fifth Assessment Report by the Intergovernmental Panel on Climate Change (AR5). The report essentially outlines how an increase of 2°C would be catastrophic [2] and provides simulation results of four likely trends called representative concentration pathways (RCP's), shown in Figure 1. Each pathway includes a prescribed set of global actions and they are used as the general road map for international environmental practices today. All pathways focus largely on reducing emissions but due to the massive amount of CO_2 already in the environment and inevitable emitters like construction and heavy transport (cargo ships, planes), also require another solution: carbon dioxide removal (CDR) systems.

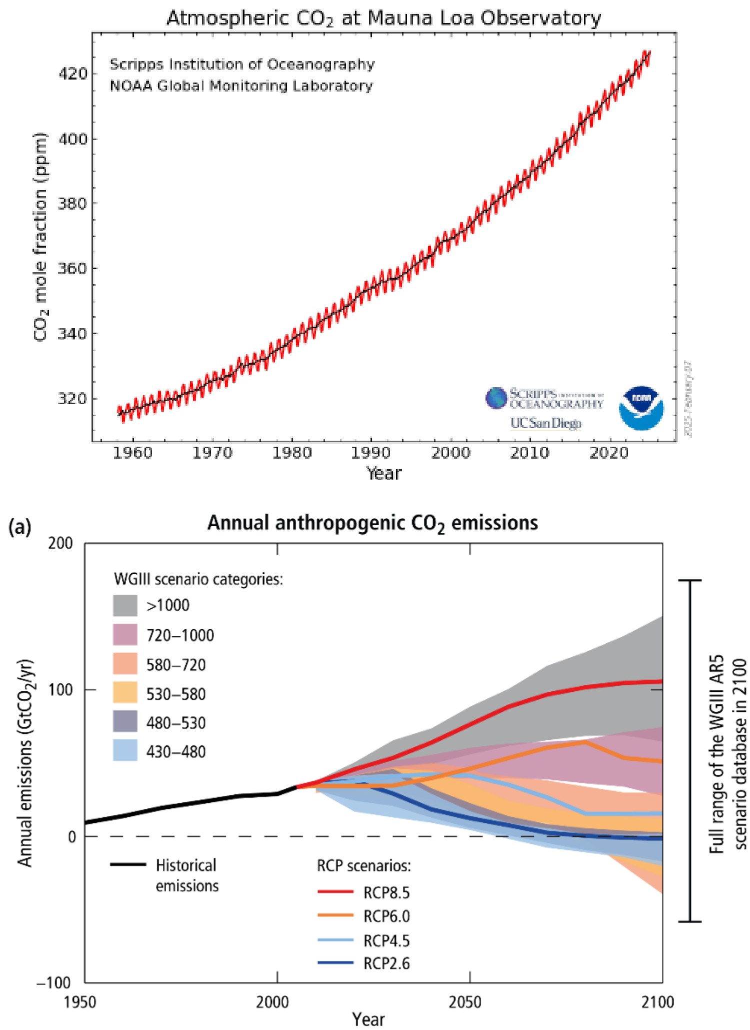

Figure 1: Top: the 'Keeling curve' which has been measuring daily temperature since 1958 in Hawaii [7]. Bottom: AR5 representative concentration pathways [9]

Also known as carbon capture and storage (CCS) or negative emission technologies, these are processes that take carbon dioxide out of the environment and store it in a solid state [2]. The captured carbon could be used to create new fuels or plastics, fertilize plants, or create dry ice, but none of these industries use CO_2 on the scale of the excess amount in the environment [3]. As a solution, many types of CDR solve this problem by solidifying the carbon onto rock in underground deposits.

The key to a successful CDR system is low energy use and high capacity, i.e., being able to capture more carbon than is released during generation of the energy used [3] and few have demonstrated this at scale. Many CDR methods like those shown in Figure 2 have been proposed, but most have yet to prove scalable for environmental CO_2. The purpose of this review is to provide an overview of the strengths, weaknesses, and technical aspects of some of the most developed CDR technologies such as point source capture, direct air capture (DAC), and direct ocean capture (DOC) and briefly touch on some other miscellaneous methods.

Approach	Capture	Storage	Description
Biomass energy and carbon capture and storage (BECCS)	Biological Indirect	Geological Pressurised	Power generation using biomass feedstocks followed by CO_2 capture from flue gases for subsequent geological storage
Direct air capture (DAC)	Non-biological Direct	Geological Pressurised	Uses chemical sorbents to absorb and then release CO_2 for subsequent geological storage
Ocean fertilisation	Biological Indirect	Biological Oceanic	Addition of nutrients to enhance growth of photosynthesising organisms, which eventually fall to the ocean floor
Ocean liming	Non-biological Indirect	Non-biological Oceanic	Calcium or magnesium oxides (lime) dissolved in sea water to increase the ocean absorption of atmospheric CO_2
Enhanced weathering	Non-biological Indirect	Non-biological Mineralised	Finely ground silicate rocks are dissolved in oceans or soils, these absorb CO_2 and are eventually sequestered in shells of organisms
Biochar	Biological Indirect	Biological Biotic	Heat treatment of biomass, locking up the carbon and can be used as a soil improver
Afforestation	Biological Indirect	Biological Biotic	Establishing and maintaining forests that have not previously been forested for a given period (e.g. 50 years) (UNFCCC, 2015)

Figure 2: Examples of existing CDR techniques [2]

2. Materials and Methods

In order to create an overview of carbon dioxide technologies, databases were accessed using the following keywords: direct air capture, economic, techno-economic assessment, and strengths and weaknesses. More than a dozen articles were identified, and those selected were found using ScholarOne Search, Elicit, EBSCOhost, and other papers citing the works found.

3. Existing Technologies

3.1 Point Source

The most effective and controllable method is called point source or flue gas capture, which removes carbon directly from combustion exhaust pipes. Factors like controllable flow properties, high concentrations, and a responsible party make exhaust flues high-impact and easy targets. Point source capture can be used to reduce emissions from transportation, building, and industrial emissions, but does not remove carbon already in the atmosphere [4].

From economic, efficacy, and feasibility standpoints, point source beats every other CDR technology. Generally, the cost to remove a pollutant from a mixture increases drastically with dilution[1], [3] and 30,000-150,000ppmCO_2 leaving flues, point source capture is far more efficient than capturing the ambient 426ppm. Carbon producers can be held responsible for checking their emissions and employing point source capture, especially since it is, by 'rule-of-thumb,' three times more energy efficient than DAC [3].

3.2 Direct Air Capture (DAC)

The most well-known type of CDR is called Direct Air Capture (DAC), which removes CO_2 from ambient air and is well described by Yafiee [6]. It works by passing air over binding chemicals, which are treated to deposit the carbon and then regenerate back into a receptive state, similar to the setup in Figure 3. These systems can be broken into three components: contactor (flow management), adsorbing/absorbing (carbon-binding chemicals), and regeneration (depositing) systems. The majority of innovation in DAC comes from the chemicals chosen to bind to the carbon, which at the low concentration of ambient air, need to be super strong-binding solvents [6]. As of 2015, there were 3 main types: aqueous solutions of strong bases (most explored), amine adsorbents, and inorganic solid sorbents [3]. A myriad of research exists into these setups that will not be discussed for brevity.

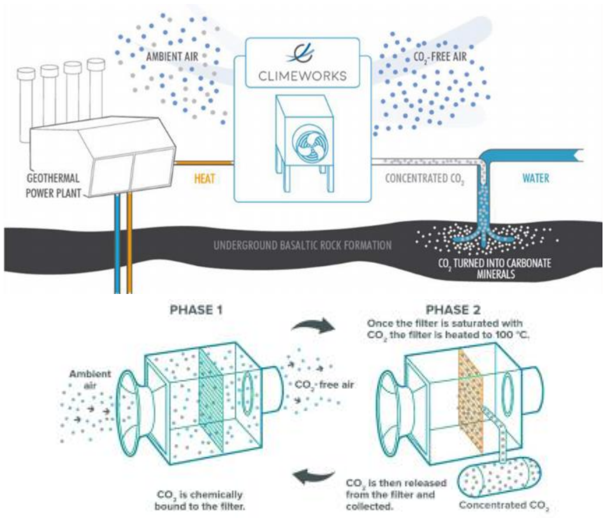

Figure 3: Diagrams of Climeworks' DAC process [5]

Being the best explored technology for CDR, researchers have been able to produce concrete prices for DAC, though research by Broehms et al. shows that they vary considerably. The driving factor in the price is the amount of energy used by a given plant design. A 2015 theoretical minimum estimated DAC would require about 1GJ/tCO_2 [1], which is about 2-4 times as much energy as point source capture [3]. However, a practical range found by Broehms et al. in 2015 estimates that plants require around 10GJ/tCO_2, while fossil fuels release about twice as much carbon. This means that renewable energy sources are crucial in net negative carbon operation [3].

DAC currently operates for about \$100-\$250USD/tCO_2 [2] but is projected to fall to \$45-\$140USD/tCO_2. A good sense of scale is that it would cost nearly \$300billion-1trillionUSD/yr to lower the global average by 1ppm/yr [3]. For scale, the most recent report from NOAA reports 426.65ppm, up from the 314.44ppm first measured in 1960 [7] and the pre-Industrial Revolution it 280ppm [1].

Assuming energy and finances were solved, DAC still faces particular challenges with respect to location and emissions. To optimize DAC success, plants should be placed in locations with favorable wind, lots of water (5-13t/CO_2)[3], and favorable air conditions. Air humidity, pressure, contaminants, and temperatures (especially below freezing) are all negative factors, but could be controlled at the expense of more energy. Factors outside of operation downstream air being dry, foggy chemical-filled, or CO_2-depleted and the amount of space (tens of thousands of square kilometers [3]) are also important considerations in design.

However, DAC has continued to prove itself increasingly successful. As of 2023, there are 19 plants (the biggest being Climeworks, Carbon Engineering, CSIRO, Global Thermostat [6]) in operation, each capturing about 10,000Mt/yr [3] at a price of \$600-\$1000USD/tCO_2. One hopeful researcher, Yafiee [6], suggests that the amount of energy used in fossil fuel refinery could be used to capture 370MtCO2/yr with present technology [6], though in order to be viable, the cost would need to drop below \$100USD/tCO_2. Generally, DAC is a mature enough technology for deployment but is

[1] 1GJ/tCO_2 = 1 gigajoule of energy to capture 1 metric tonne of carbon dioxide

[2] 1 USD/tCO_2 = 1 United States dollar to capture and store 1 metric tonne of carbon dioxide

[3] 1Mt/yr = 1 million metric tonnes of CO_2 captured per year

hindered by support and resources for such large infrastructure.

3.3 Direct Ocean Capture (DOC)

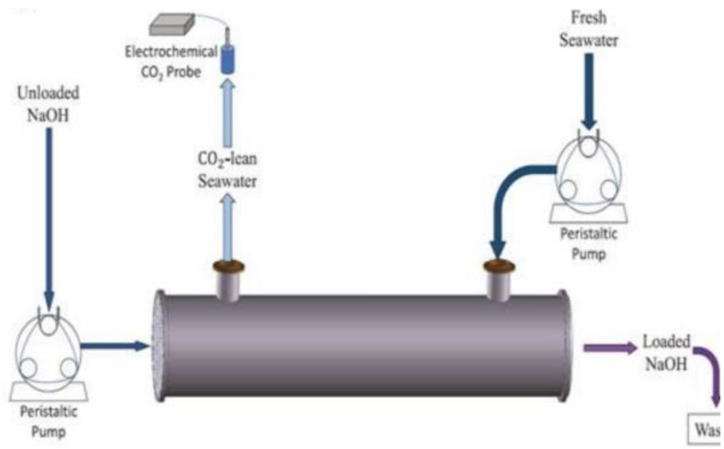

Figure 4: Sample diagram of DOC seawater treatment [6]

Another technology-infrastructure based CDR system is called Direct Ocean Capture (DOC), in which an offshore or coastal plant removes dissolved carbon dioxide from seawater as in Figure 4. As Yafiee brings up, “oceans are essential to the global carbon cycle due to their vast carbon storage capacity, of approximat[ely] 38,000 gigatons, which is significantly greater than either the 700 gigatons capacity of the atmosphere or the 2,000 gigatons capacity of the terrestrial biosphere [6].” Similar to the increase in atmospheric CO_2 [7], regular measurements have shown steady increase in ocean acidity [6]. Many marine species, especially those with calcified exoskeletons, like coral and crustaceans, are sensitive to pH changes.

DOC is one of a few CDR techniques that focuses on extracting carbon from the ocean while also directly addresses its acidity. Oceans typically absorb 27% of CO_2 emissions and convert them into bicarbonate and carbonate ions, which slowly acidifies all fresh and saltwater bodies [6]. DOC is a process that converts carbonate CO_3^{1-} and HCO_3^{2-} ions back into carbon dioxide via electrodialysis (electrically forcing ions through membranes), passive membranes, or chemical scrubbers. However, these systems are still in research phases and there is no strong development in any one technique [6]. Theoretical calculations for some electrochemical DOC processes appear to be slightly more efficient (1.3kWh/kg vs 1.54-2.45 kWh/kg) than some commercial DAC systems, and improvements in chemicals, materials, and engineering could improve the margin. However, the current state of the technology is not cost-viable.

The benefits of DOC are still predictions based on pilot plants but appear promising. Though they sit at about twice as expensive as leading DAC plants, they have other advantages. DOC plants can operate anywhere with ocean water, an energy source, and CO_2 storage access. Thie means that they could even be built offshore, taking up no land and eliminating transportation costs by depositing directly to undersea storage locations.

DOC is theoretically more efficient, cheaper, better suited to small scale than DAC, but far less explored. The only notable pilot plant, created by Captura, removes only 100 tons/yr from Newport Beach, CA. The technology for DOC is still in its growth period and could provide significant results but currently is not ready for use. As DAC paves the way with work on materials, engineering, and public reception, DOC and other CDR technologies will have an easier path forward.

3.4 Other Techniques

Many other techniques are still in theoretical or research phases and vary widely in application, method, and final destination of the carbon. Some of the more mature systems include biomass energy and carbon capture and storage (BECCS) and afforestation. BECCS, which is more infrastructure process than technology, describes energy generation through combustion of biomass fuel with point-source capture at the flue. Effectively, it captures carbon dioxide through photosynthesis, then captures it again for sequestration after burning [2]. In afforestation, land is converted to forests, which, like BECCS involves a whole complex of land-use considerations [8]. One powerful tool in researching land-use effects is a spatially and temporally discretized simulation that accounts for economics, food production, emissions, water, and human population called MAgPIE4.0 (Model of Agricultural Production and its Impact on the Environment). Produced by the Potsdam Institute for Climate Impact Research, researchers have been able to use it to recommend certain strategies for CDR methods, such as deciding to implement BECCS only when carbon taxes are high [4], [8].

Generally, other systems for CDR are less developed and have no large-scale quantitative data about feasibility, cost, efficiency, or other logistics. Nevertheless, the last decade has shown significant growth in both the number and maturity of CDR technologies. In the next few years, if not already, these technologies will be far better explored and more likely to reach a commercial scale.

4. Conclusion

With the exception of DAC and point source capture, CDR technologies are generally underdeveloped for full-scale deployment but have promising features. While DAC appears to be the most developed technique for CDR, it still falls short of point source capture on simplicity, cost, and many other logistics. Undoubtedly, minimizing climate change and following global environmental pathways will require a combination of both these and other techniques like DOC and BECCS, and every net impact makes a difference. However, the key in combatting climate change is ensuring that as these technologies mature, they are supported and applied globally and equitably.

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